

## Activity 6 Pre-Lab — Dehydration of Cyclohexanol

[Your name] & [partner]

[MM/DD/YY]

Organic Lab · Synthesis of cyclohexene + purification by simple distillation

■ **Black = write this in your notebook** ■ **Blue = context, don't copy it** Copy the header onto every page. Draw anything in a dashed box yourself.

### Purpose

Synthesize **cyclohexene** by the acid-catalyzed **dehydration (E1)** of **cyclohexanol** using 85% phosphoric acid, isolate it by **simple distillation**, then check its purity/identity by TLC, the bromine test, and IR.

#### 🔥 DRAW — the reaction

input chemical structure of cyclohexanol  $\rightarrow$  cyclohexene +  $\text{H}_2\text{O}$  over the arrow write: 85%  $\text{H}_3\text{PO}_4$ , heat (distill) · mechanism = E1 (protonate OH  $\rightarrow$  lose  $\text{H}_2\text{O} \rightarrow$  carbocation  $\rightarrow$  lose  $\beta\text{-H}$ )

### Chemical Reagents & Safety Table

**CONTEXT** ▶ Required: cyclohexanol, phosphoric acid, cyclohexene, sodium sulfate, sodium chloride. Typical SDS values — cross-check the SDS tile on Labflow.

| Chemical                   | Role in lab                       | MW     | Amount used                 | Physical properties                                            | Hazards (GHS)                                                                                                               |
|----------------------------|-----------------------------------|--------|-----------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Cyclohexanol               | Starting material (limiting)      | 100.16 | 10 mL (~9.6 g, ~0.096 mol)  | Colorless liquid/waxy solid; bp 161 °C; d 0.96 g/mL; mp ~25 °C | H302 harmful if swallowed · H315 skin irritation · H332 harmful if inhaled · H335 respiratory irritation                    |
| Phosphoric acid (85%)      | Acid catalyst / dehydrating agent | 98.00  | 12 mL                       | Colorless viscous liquid; d 1.69 g/mL                          | H290 corrosive to metals · H314 causes severe skin burns & eye damage                                                       |
| Cyclohexene                | Product                           | 82.14  | product (theor. ~0.096 mol) | Colorless liquid; bp 83 °C; d 0.81 g/mL; flash pt -12 °C       | H225 highly flammable · H302 harmful if swallowed · H304 may be fatal if swallowed/enters airways · H315 · H411 aquatic tox |
| Sodium sulfate (anhydrous) | Drying agent                      | 142.04 | pea-sized                   | White solid; hygroscopic                                       | No significant GHS hazards (drying agent) — still avoid dust/eyes                                                           |
| Sodium chloride (sat. aq.) | Brine wash (removes water)        | 58.44  | 2 × 10 mL                   | Colorless aqueous solution                                     | No significant GHS hazards                                                                                                  |

**FYI** ▶ also used: 10%  $\text{Na}_2\text{CO}_3(\text{aq})$  to neutralize phosphoric acid that distills over (fizzes  $\rightarrow$   $\text{CO}_2$ , vent often!); charcoal boiling chips (regular stones dissolve in  $\text{H}_3\text{PO}_4$ ).

**GRAB FIRST:** 100-mL round-bottom (reactor) · 50-mL round-bottom (receiver) · simple-distillation head + condenser + thermometer adapter + thermometer · heating mantle + Variac · lab jack + clamps · ice-water bath · small separatory funnel · 25-mL Erlenmeyer · charcoal boiling chips.

### Procedure Plan (left = your plan, right = fill in during lab)

#### Plan (write before lab)

##### Part 1 — Synthesis + distillation

1. Measure **~10 mL cyclohexanol** into the **100-mL RBF**; record exact volume.
2. Add **12 mL 85%  $\text{H}_3\text{PO}_4$**  + a few **charcoal boiling chips**, swirl to mix.  
 *$\text{H}_3\text{PO}_4$  is corrosive — gloves; follow instructor's handling.*
3. Clamp the reactor to the bars; build a **simple-distillation** apparatus (still head → thermometer → condenser → 50-mL receiver in an **ice-water bath**). Heating mantle on a **raised lab jack** below. *Jack must drop enough to slide the hot mantle out.*
4. Instructor checks setup → heat with the **Variac**.
5. Keep the **vapor temp  $\leq 83^\circ\text{C}$**  (adjust Variac / lower mantle). Record the steady distilling temp.  *$83^\circ\text{C}$  = cyclohexene bp; leaves cyclohexanol +  $\text{H}_3\text{PO}_4$  behind.*
6. Stop (lower jack, mantle off) when **5–10 mL** remain in the reactor. **Never distill to dryness.**
7. Neutralize: slowly add **10%  $\text{Na}_2\text{CO}_3(\text{aq})$**  to the receiver in 5-mL portions, swirl till fizzing stops → transfer to sep funnel. *Vent often in the hood —  $\text{CO}_2$  builds pressure.*
8. Wash the organic layer **2 × 10 mL saturated NaCl**; drain off the aqueous each time. *Product = organic (top) layer.*
9. Transfer product to a 25-mL Erlenmeyer; dry over a **pea-sized** amount of anhydrous  **$\text{Na}_2\text{SO}_4$**  until no longer cloudy.
10. Decant the clear cyclohexene into a pre-weighed flask; weigh → mass of product → % yield.

##### Tests (do all before discarding product)

11. **TLC** (30% EtOAc/hexanes) · **Bromine test** ( $\text{Br}_2$  decolorizes → alkene) · **IR** (look for  $\text{=C-H}$  ~3050,  $\text{C=C}$  ~1650; loss of broad  $\text{O-H}$ ).

#### ✍️ WRITE

$$\% \text{ yield} = (\text{actual mol cyclohexene} \div \text{theoretical mol}) \times 100$$

#### Experiment Notes (in lab)

Exact cyclohexanol volume: \_\_\_\_\_ mL

Steady vapor temp: \_\_\_\_\_  $^\circ\text{C}$

Distillation observations:

Workup — layers / venting / bubbling:

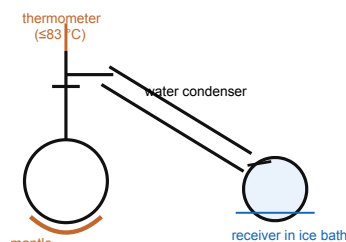
Product appearance (clear? turbid?):

Mass flask: \_\_\_\_\_ g flask+product: \_\_\_\_\_ g

Mass product: \_\_\_\_\_ g % yield: \_\_\_\_\_

Bromine test / TLC  $R_f$  / IR peaks:

### Setup — draw this



simple distillation

#### CONTEXT (don't copy) ▶

**Why  $83^\circ\text{C}$  cap?** cyclohexene boils at  $83^\circ\text{C}$ ; cyclohexanol ( $161^\circ\text{C}$ ) and  $\text{H}_3\text{PO}_4$  stay behind, so the distillate is mostly pure product.

**Why brine +  $\text{Na}_2\text{SO}_4$ ?** saturated NaCl pulls water out of the organic layer; anhydrous  $\text{Na}_2\text{SO}_4$  then dries the last traces (cloudy → clear).

### Clean-up & Conclusions

- Acidic pot residue → neutralize/dispose per instructor; organic waste for cyclohexene after tests; aqueous washes → aqueous waste; return glassware; do your bench task.
- **Conclusions** (~4 sentences [after lab](#)): your % yield and a reason it wasn't 100% (product left in pot, evaporation, incomplete rxn); what the bromine test + IR  $\text{C=C}$  confirmed; and how distillation purified the product.

